

Combining Linear-Fee Market Makers

The infimal convolution of cost-function makers with proportional fees: spreads, sparse routing, and a derived order book

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Working draft v0.1 · 2026

Abstract

A cost-function market maker charging a proportional fee $f|s|$ per fill quotes, after Fenchel conjugation, exactly a bid-ask spread: the conjugate of the fee-bearing cost is the no-fee conjugate evaluated at a soft-thresholded price, zero on a band of half-width f around the marginal price. Combining n such makers is the infimal convolution of their effective costs, and the optimal split equalises fee-adjusted marginal prices at a single clearing price, found by a monotone scalar root-find; any maker whose quote band contains the clearing price receives exactly zero flow, so the split is sparse in the manner of lasso solutions. The aggregate supply curve is a consolidated limit order book whose levels are smooth curves. The identities are classical and the near misses are catalogued; the assembly, fees inside the costs rather than beside them, appears unoccupied, and it leaves each maker free to quote its own fee with routing as the discipline.

1 Setup

A single risk settles in $[-1, 1]$. Maker $i \in \{1, \dots, n\}$ posts a closed, finite, differentiable, strictly convex cost $C_i : \mathbb{R} \rightarrow \mathbb{R}$, holds inventory q_i , and charges a proportional fee $f_i \geq 0$, so a fill s (signed) costs the trader

$$\tilde{C}_i(s) = C_i(q_i + s) - C_i(q_i) + f_i|s|.$$

Without the fee this is the standard cost-function maker ([Hanson 2003](#); [Chen and Pennock 2007](#); [Abernethy et al. 2013](#)), whose charge telescopes over any path returning to the same state: round trips are free, so uninformed flow pays nothing and adverse selection cannot be recovered from volume. That no state-dependent cost can charge a round trip, and that a path-dependent volume charge repairs it, is due to Othman and Sandholm ([2012](#)); the linear charge $f_i|s|$ is their volume levy in its simplest convex form, chosen here because it conjugates in closed form. Write $g_i(s) = C_i(q_i + s) - C_i(q_i)$ and $m_i = C'_i(q_i)$ for the marginal price.

2 A linear fee is a bid-ask spread

Lemma 1 (fee-spread duality). *Let $T_f(x) = \text{sign}(x) \max(|x| - f, 0)$ denote the soft-threshold. Then*

$$\tilde{C}_i^*(p) = g_i^*(m_i + T_{f_i}(p - m_i)),$$

and in particular $\tilde{C}_i^*(p) = 0$ if and only if $|p - m_i| \leq f_i$.

Proof. The conjugate of a sum of closed proper convex functions with overlapping relative interiors of domains is the infimal convolution of the conjugates (Rockafellar 1970, Thm. 16.4), and the conjugate of $f|\cdot|$ is the indicator of $[-f, f]$. Hence $\tilde{C}_i^*(p) = \min_{|u| \leq f_i} g_i^*(p - u)$, the minimum of the convex function g_i^* over the interval $[p - f_i, p + f_i]$. Since $g_i^* \geq 0$ with equality exactly at m_i , the minimum is attained at the projection of m_i onto the interval, which is $m_i + T_{f_i}(p - m_i)$. ■

$g_i^*(p)$ is the profit extractable from maker i at external price p (in producer-theory terms the profit function, with supply its derivative by Hotelling's lemma (Hotelling 1932)). The lemma says the fee-bearing maker quotes $\text{ask}_i = m_i + f_i$ and $\text{bid}_i = m_i - f_i$ and is untouchable in between: a proportional fee is not like a spread, it is one, the same duality by which a proportional transaction cost confines the pricing functional to the bid-ask band (Jouini and Kallal 1995).

3 Combining makers

Theorem 2 (combination). Let $\tilde{C} = \tilde{C}_1 \square \dots \square \tilde{C}_n$ and define each maker's supply

$$s_i(p) = \begin{cases} (C'_i)^{-1}(p - f_i) - q_i, & p \geq \text{ask}_i, \\ 0, & \text{bid}_i < p < \text{ask}_i, \\ (C'_i)^{-1}(p + f_i) - q_i, & p \leq \text{bid}_i, \end{cases}$$

each non-decreasing in p . Fix a demand Δ for which a clearing price p^* with $\sum_i s_i(p^*) = \Delta$ exists. Then:

- (i) $\tilde{C}^* = \sum_i \tilde{C}_i^*$, a sum of soft-thresholded profit functions;
- (ii) the split $s_i = s_i(p^*)$ attains $\tilde{C}(\Delta)$, and any optimal split satisfies $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$ for $s_i \neq 0$ and $|p^* - m_i| \leq f_i$ for $s_i = 0$;
- (iii) the split is sparse: every maker whose quote band strictly contains p^* trades exactly zero.

Proof. (i) is the conjugate-sum identity applied to the convolution (Rockafellar 1970). For (ii), the split is feasible by choice of p^* , and $p^* \in \partial \tilde{C}_i(s_i(p^*))$ for every i : when $s_i(p^*) \neq 0$ the subgradient is $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$, and when $s_i(p^*) = 0$ it is the interval $[m_i - f_i, m_i + f_i] \ni p^*$. A common multiplier certifying every coordinate is exactly the optimality condition for $\min\{\sum_i \tilde{C}_i(s_i) : \sum_i s_i = \Delta\}$. (iii) restates the zero branch. ■

The computation is a scalar monotone root-find whatever n is, and the fee costs nothing beyond a horizontal shift of each supply curve. The $|s|$ terms are an ℓ_1 penalty, so sparsity arrives for the same reason it does in the lasso, and for the same reason proportional transaction costs produce no-trade regions and sparse portfolios (Olivares-Nadal and DeMiguel 2018). A small trade routes entirely to the tightest quote; a growing trade pushes that maker's fee-adjusted marginal price through the next band and spills over, consuming makers in fee order.

Corollary 3 (zero fees). With $f_i \equiv 0$ the convolution reduces to the fee-free merge: conjugate regularisers add, and for a perspective family $C_b(q) = b C_1(q/b)$ liquidity adds, $C_{b_1} \square C_{b_2} = C_{b_1+b_2}$. This is the merge law of the companion composition paper and of Bhaskara et al. (2023).

Corollary 4 (the order book). The aggregate supply $S(p) = \sum_i s_i(p)$ is non-decreasing, identically zero on $(\max_i \text{bid}_i, \min_i \text{ask}_i)$ when that interval is non-empty, flat wherever every maker's band covers p , and smooth and strictly increasing wherever some maker is in the money. Read as a market: best bid and ask are the tightest quotes, depth at each price is the sum of the active makers'

closed-form supplies, and large orders walk the levels. The aggregate of linear-fee makers is a consolidated limit order book, and in producer-theory terms Theorem 2 is Marshall’s horizontal summation of firm supply curves (Marshall 1890; Mas-Colell et al. 1995) with the reversibility of share production patched by the fee.

4 Discussion

Nothing requires the fees to be administered. Each maker may quote its own f_i : a quote inside the aggregate spread earns nothing, a quote too tight is picked off by informed flow, and the undercutting happens inside the same minimisation that clears the trade, since the inf in the convolution is a minimum over quotes. The surviving fee is the competitive adverse-selection charge of classical microstructure (Glosten and Milgrom 1985; Biais et al. 2000), reached here through routing rather than through a dealer game.

The ingredients are published. Path-dependent profit charging is Othman and Sandholm (2012); aggregation of cost functions by infimal convolution is Bhaskara et al. (2023), with the geometric reading in Angeris et al. (2024); clearing-price routing across fee-bearing venues, with zero flow to venues whose band contains the dual price, is Angeris et al. (2022) and Diamandis et al. (2023); the fee-induced no-trade region is Milionis et al. (2024); the order book assembled from competing liquidity suppliers is Glosten (1994), with convergence of strategic schedules to it in Biais et al. (2000); frictions inside inf-convolution risk sharing appear in Ludkovski and Young (2009). What was not found in any of these is the present packaging: the fee folded into the cost so that one conjugate identity (Lemma 1) yields the spread, the sparsity, the scalar clearing computation, and the book, with per-maker fees as free competitive variables. A fuller per-claim assessment, with the searches that ground it, is in the repository note [proportional-fees-and-the-order-book.md](#); a reference implementation with theorem tests is in [fee_routing.py](#).

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